Name: ______________________

UHID: ______________________

Homework
#05: RAM & ROM

Directives

1. Write the **directive** to create a **32-bit** <u>constant</u> named **const1** initialized to **1**.

2. Write the **directive** to create a **16-bit** <u>constant</u> named **const2** initialized to **0x2**.

3. Write the **directive** to create an **8-bit** <u>constant</u> named **const3** initialized to **2_11** (binary 3).

4. Write the **directive** to create a **32-bit** <u>variable</u> named **var4**.

5. Write the **directive** to create a **16-bit** <u>variable</u> named **var5**.

6. Write the **directive** to create an **8-bit** <u>variable</u> named **var6**.

Load/Store Registers from/into RAM/ROM

7. Write the assembly code to **load** register **R0** with the value of the **32-bit** constant named **const1**.

8. Write the assembly code to **load** register **R0** with the value of the **16-bit** constant named **const2**.

9. Write the assembly code to **store** the value in register **R0** to the **32-bit** variable named **var4**.

10. Write the assembly code to **store** the value in register **R0** to the **8-bit** variable named **var6**.

Programming

11. Write the assembly program to accomplish the following C instruction: (Both const1 and var4 are 32 bits and they already defined and initialized.)

```
Var4 = const1;
```

12. Write the assembly program to accomplish the following C code: (varY is 32 bits and already defined and initialized.

```
While(varY<10) {

     varY++;
}
```